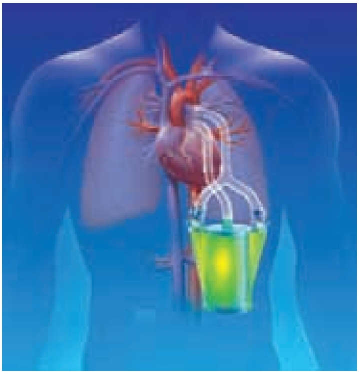
instance, a genetic test can tell you that you are at high risk for Huntington's disease, but you cannot do much to stop the progression of this incurable neurological disease. There is a consensus, however, that one of the biggest impacts of nanotechnology will be rapid DNA sequencing. New nanopore devices are poised to make complete genome sequencing rapid, cheap and widely available. That means it will be possible for individuals to find out the sequences of all their genes, including those linked to disease. However it remains to be seen whether or not the public will embrace personal genomic sequencing, and whether or not it might pave the way for predictive medicine or derail it.

2. **Preemptive medicine -** This vision focuses more on early intervention, to help doctors detect treatable diseases earlier so that they can help patients preempt the full-blown development of illness or at least manage it effectively over a lifetime. Nanotechnology can enhance the development of more- sensitive diagnostic tests, as well as devices for health monitoring and disease management. Diabetes care is one obvious and important area that would stand to benefit from it. Consider that if new, nano-based diagnostic tests could detect the earliest stages of insulin resistance; one could make changes in one's diet and exercise to slow the progression of the disease. A health-monitoring device at home could help one maintain one's regimen by providing positive feedback on your blood sugar levels. It would also help one manage the dosage and timing of insulin intake more effectively. Similarly, preemptive medicine might be used to help a large portion of the population more effectively deal with cardiovascular disease and hypertension. This model could be expanded and used to manage many chronic diseases—say, for example, lupus or arthritis— as new diagnostic tests, monitoring capabilities and therapies became available.


Never seen before cures,

3. **Personalized medicine -** This vision aims to make medicine more personalized by using information about an individual, to specifically tailor his or her treatment. A doctor has a much greater chance of coming up with an effective medical strategy if he or she knows something about a patient's disease subtype, metabolism (particularly as it relates to drugs), liver status and risk